

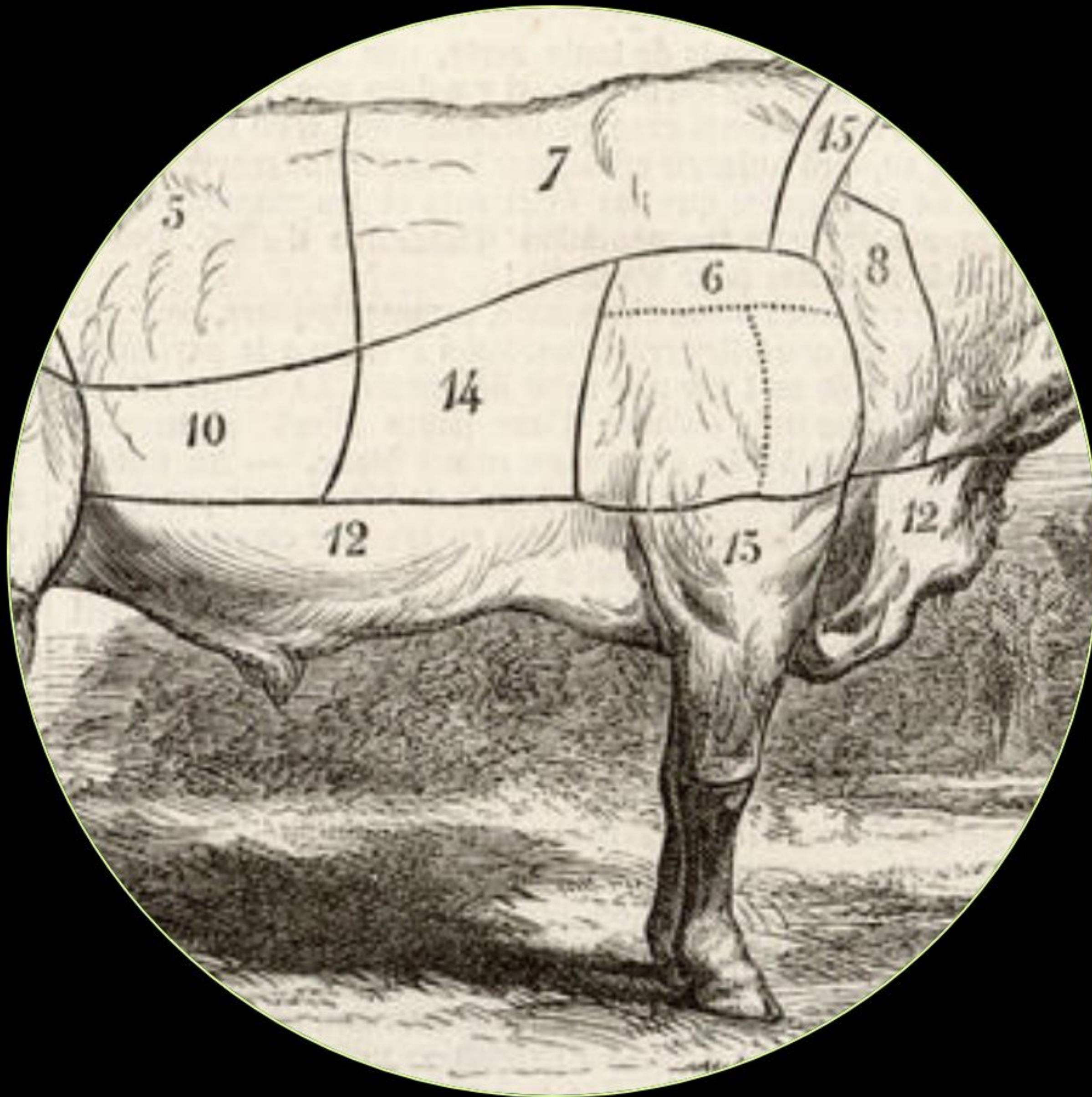
The Wisdom of the Crowd & Ensemble Learning

Harnessing Collective Intelligence in Human and Machine Systems

Part 1: The Human Crowd

Why groups are smarter than individuals

Galton's Ox: The 1906 Experiment



The Statistical Miracle

Sir Francis Galton observed 800 villagers guessing the weight of an ox. While individual guesses were wildly inaccurate, the **mean estimate** was stunningly precise.

- Crowd Mean: **1,197 lbs**
- Actual Weight: **1,198 lbs**

The collective error was less than 0.1%.

4 Conditions for a "Wise" Crowd



Diversity

Each person must have some private information.



Independence

Opinions are not influenced by others.



Decentralization

People can draw on local knowledge.



Aggregation

A mechanism to turn private judgments into a collective decision.

Real-World Crowd Intelligence

Beyond the Jellybean Jar

- **Ask the Audience:** Statistically correct 91% of the time, vs. 65% for individual "experts."
- **Prediction Markets:** IEM market prices outperform polls 74% of the time.
- **Google Search:** Hyperlinks act as individual "votes" for a page's relevance.



Part 2: Machine Learning Ensembles

Turning "Weak Learners" into "Strong Models"

Why Ensemble Techniques Work

Variance Reduction

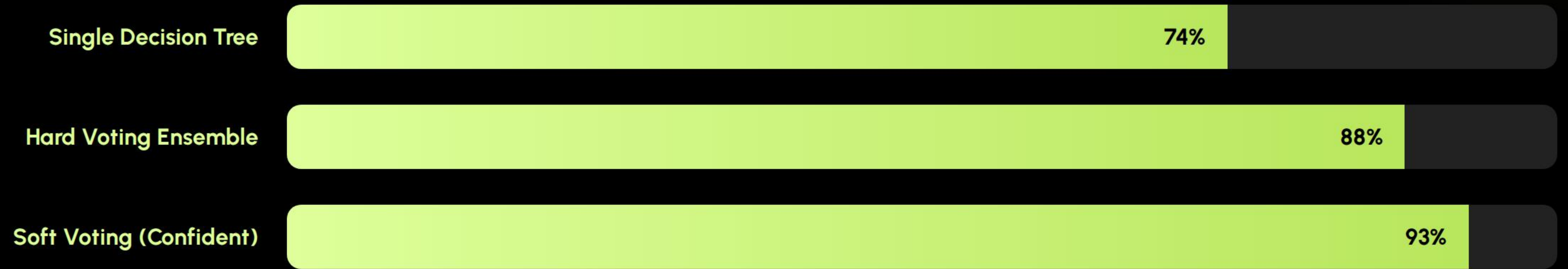
By averaging many noisy models, we smooth out idiosyncratic errors and prevent **overfitting**.

Bias Mitigation

Combining models with different biases allows the ensemble to capture complex patterns a single model misses.

$$E [\text{Ensemble}] = \frac{1}{N} \sum_{i=1}^N f_i (x)$$

Performance: Single vs. Ensemble



Soft voting usually outperforms hard voting as it considers the **probability/confidence** of each model.

Bagging & Random Forests

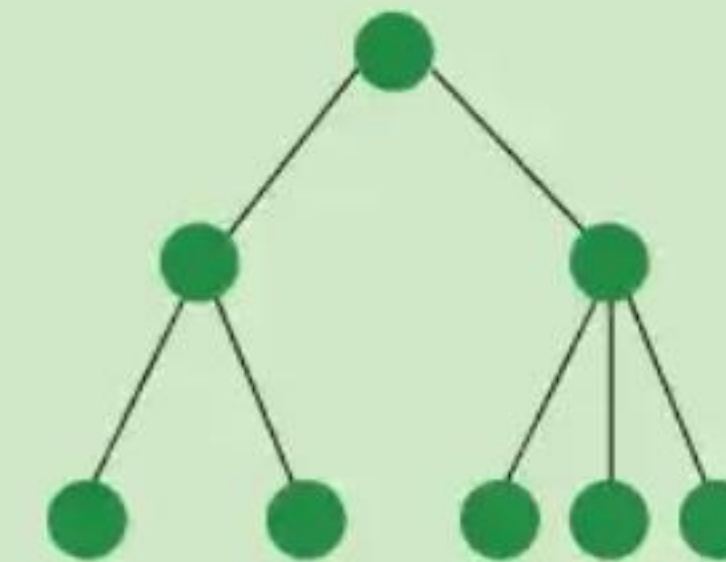
Bootstrap Aggregating

Bagging trains multiple models in **parallel** on random subsets of data (with replacement).

- **Random Forest:** Builds hundreds of decision trees using random features.
- Reduces variance significantly.
- Robust against noise in the training set.

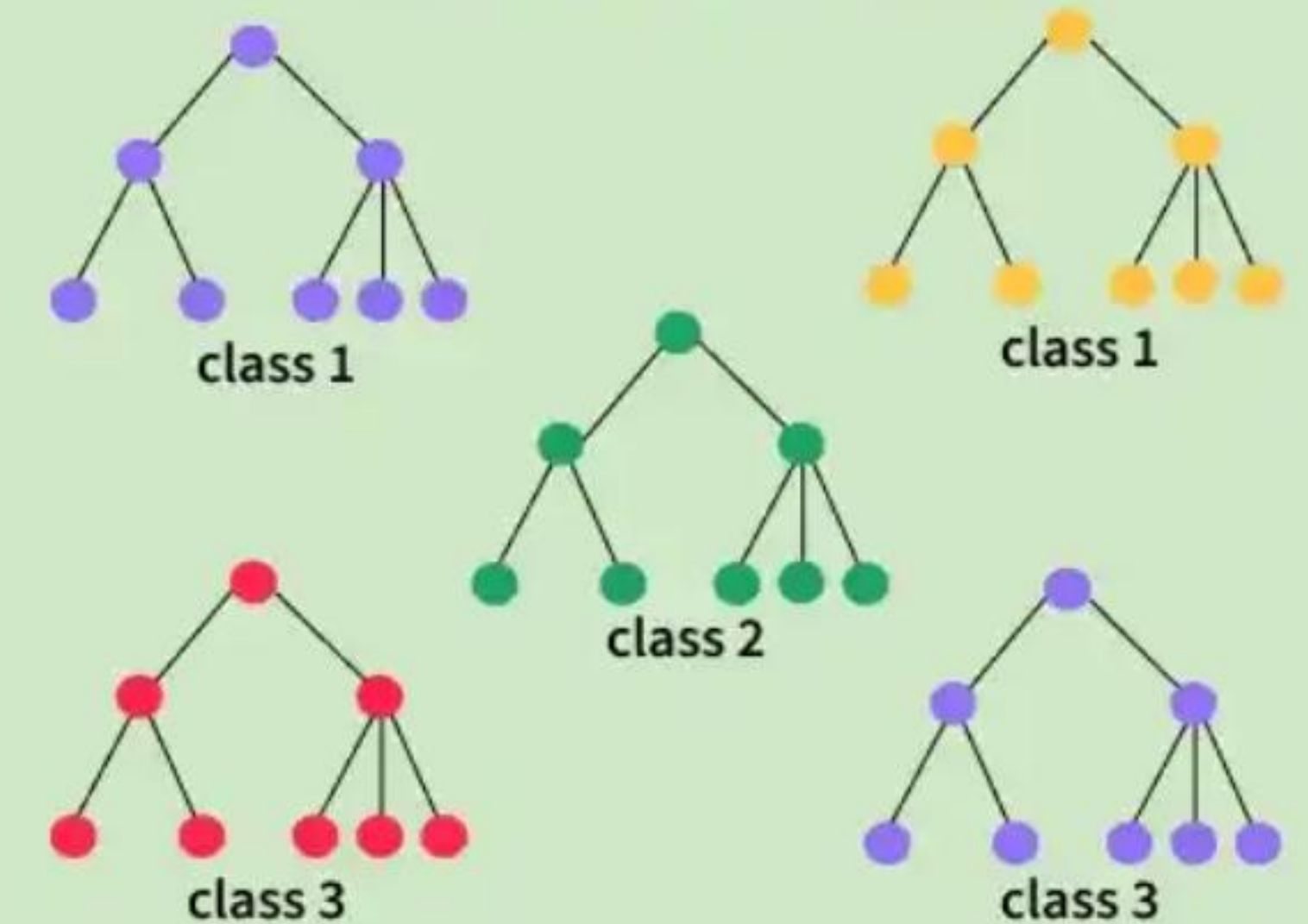
Single Decision Tree

Prediction from a single decision path



Random Forest

Ensemble of trees for more accurate and robust prediction



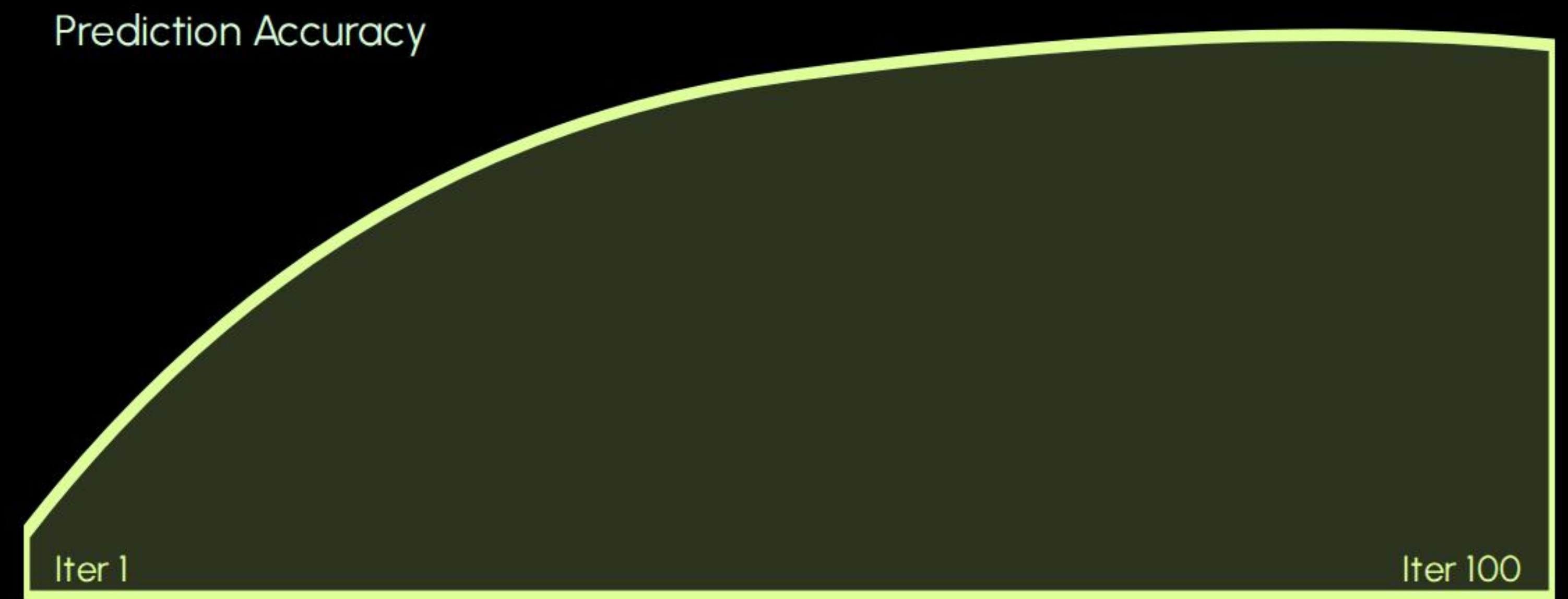
Boosting: Sequential Learning

Learning from Mistakes

Boosting builds models **sequentially**. Each new model focuses on instances the previous ones misclassified.

Popular Algorithms:

- **AdaBoost**: Adaptive Boosting
- **XGBoost**: Extreme Gradient Boosting



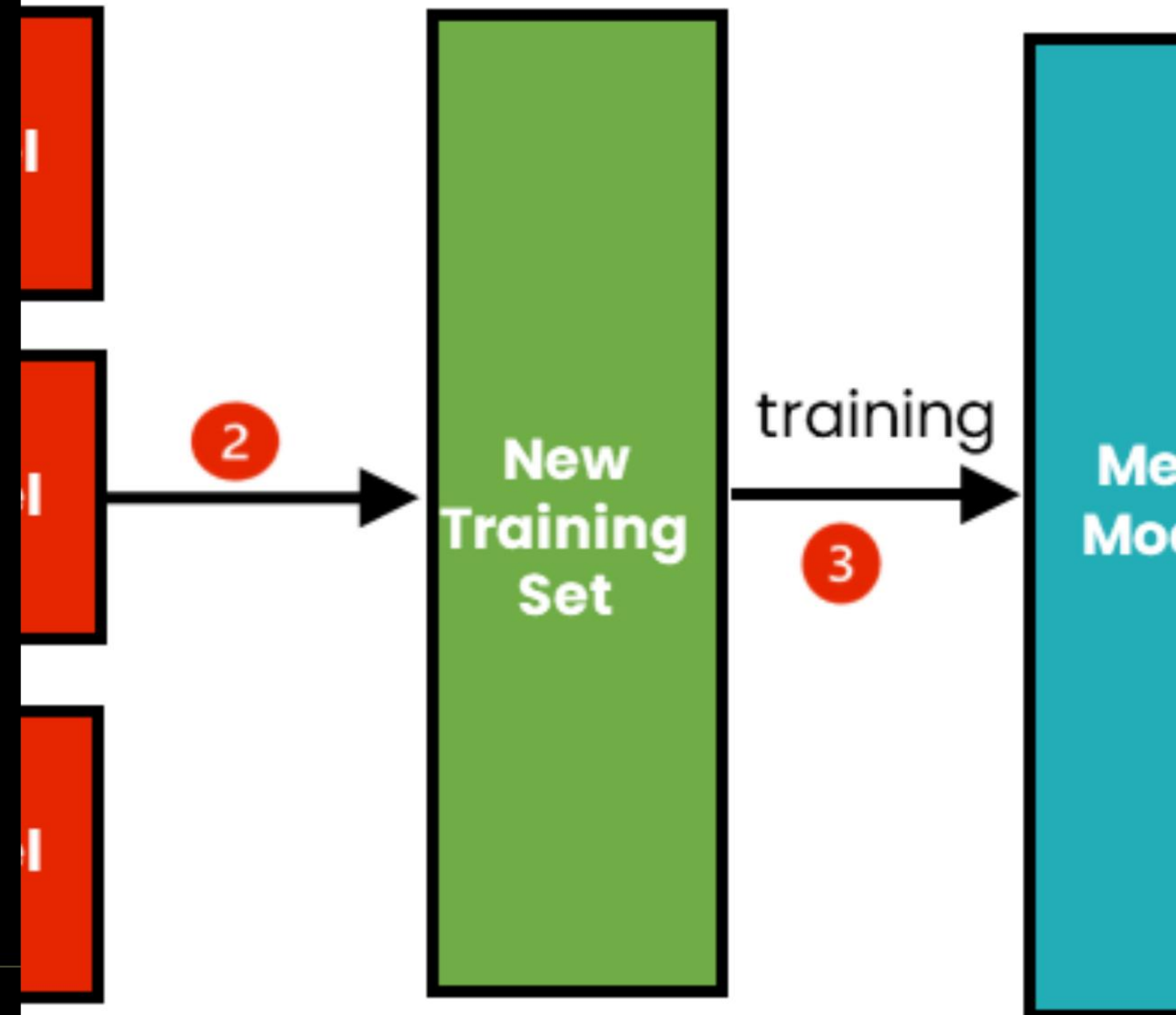
The Process of Stacking

Stacking & Meta-Models

The Board of Directors

Unlike simple voting, **Stacking** uses a "Meta-Model" to learn how to best combine the predictions of the base models.

The Meta-Model figures out which base models are more reliable for specific data regions, acting as a "blender."



Questions?

Collective intelligence leads to superior outcomes.

Key Takeaways:

- ✓ Errors cancel out in diverse groups.
 - ✓ Bagging reduces variance (Parallel).
 - ✓ Boosting reduces bias (Sequential).
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Image Sources



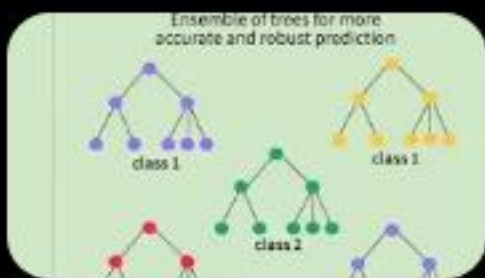
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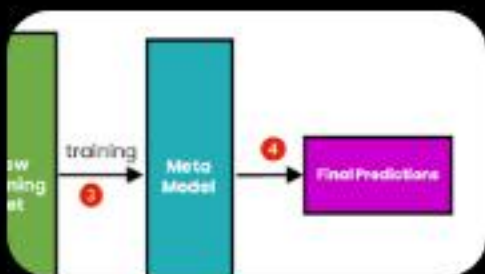
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